

MAXIMIZING THE POTENTIAL OF SYNTHETIC DATA: INSIGHTS FROM RANDOM MATRIX THEORY

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ABSTRACT

Synthetic data has gained attention for training large language models, but poor-quality data can harm performance (see, e.g., Shumailov et al. (2023); Seddik et al. (2024)). A potential solution is data pruning, which retains only high-quality data based on a score function (human or machine feedback). Previous work Feng et al. (2024) analyzed models trained on synthetic data as sample size increases.

Using random matrix theory, we generalize this analysis and derive the performance of a binary classifier trained on a mix of real and pruned synthetic data in a high dimensional setting. Our findings identify conditions where synthetic data could improve performance, focusing on the quality of the generative model and verification strategy. We also show a smooth phase transition in synthetic label noise, contrasting with prior works on sharp transition in infinite sample limits. Our extensive experimental setup validates our theoretical results.

1 INTRODUCTION

The landscape of large language models (LLMs) is evolving rapidly, with a growing trend towards training models on a combination of real and synthetic data. This synthetic data is often generated by previously trained models (Allal et al., 2024; Ben Allal et al., 2024; Abdin et al., 2024). However, the quality of these generators can significantly impact the performance of newly trained models, potentially leading to model collapse (Shumailov et al., 2023), a phenomenon in which the model drastically degrades in performance.

Model collapse has been extensively studied, both empirically (Guo et al., 2023) and theoretically (Seddik et al., 2024), highlighting the potential risks associated with training on synthetic data. To mitigate these risks, researchers have proposed various strategies, including the verification of AI-synthesized data (Feng et al., 2024). This approach aligns with the widely adopted Reinforcement Learning from Human Feedback (RLHF) technique (Kaufmann et al., 2023). Feng et al. (2024) provided theoretical support for this strategy by analyzing synthetic data as Gaussian mixtures with noisy labels, using linear binary classifiers and scalar parameters to control verifier quality. Their findings reveal a sharp performance transition: *under infinite synthetic sample size conditions, model accuracy shifts from zero accuracy (due to errors in synthetic data and verification) to optimal performance as these errors decrease.*

While current theoretical studies primarily focus on label noise in synthetic data (Dohmatob et al., 2024; Gerstgrasser et al., 2024; Feng et al., 2024), they often overlook potential distribution shifts in the feature space between real and synthetic data. This gap is particularly relevant in practical scenarios where generative models are trained on finite real data sets, potentially leading to imperfect learning of the underlying distribution.

Our paper addresses this gap by proposing a statistical model that accounts for both distribution shifts in the feature space and label noise. In our model, we induce distribution shifts in the feature space by supposing that the statistics of synthetic data are empirical estimates of the underlying real data statistics. In a finite sample size regime, these estimates may be biased, resulting in distribution

shifts between real and synthetic data. Leveraging random matrix theory, we derive the theoretical performance of a binary classifier trained on a mixture of real and pruned (i.e., verified) synthetic data in a high-dimensional setting. Our analysis provides conditions under which synthetic data improves performance, emphasizing the critical roles of the generative model’s quality and the efficacy of the synthetic data verification strategy. Lastly, we show that the sharp phase transition phenomenon identified in (Feng et al., 2024) in the infinite sample size limit is a particular case of a general result, where smooth phase transitions can take place.

Summary of contributions. Our contributions are four fold:

- We introduce a statistical model for studying synthetic data that accounts for label and feature noise, extending beyond previous models that only consider label noise.
- By leveraging random matrix theory, we characterize the performance of a binary classifier trained on a mixture of real and synthetic data in a high-dimensional setting.
- When training only on synthetic data, we find a smooth phase transition in classifier performance, generalizing the work of Feng et al. (2024) on sharp transitions in infinite sample size limit.
- We validate our results with extensive experiments (toy example and realistic LLM setups).

2 RELATED WORK

The use of synthesized data for model training has gained significant traction in recent years, particularly with the widespread adoption of large language models (LLMs) that rely on large amounts of data in their training stages. Several studies have explored the impact of synthesized data on model performance, revealing both its advantages and limitations. A primary concern is the phenomenon of model collapse Shumailov et al. (2023), where the iterative use of generated data for model training results in a degradation of model quality. This issue has been explored theoretically and empirically across multiple studies (e.g. Alemohammad et al. (2023); LeBrun et al. (2021); Bohacek & Farid (2023); Seddik et al. (2024); Dohmatob et al. (2024)).

Seddik et al. (2024) investigated model collapse in recursive training settings, where new models are trained on data generated by previous models. They demonstrate that recursive training on purely synthetic data inevitably leads to performance degradation. However, they show that mixing real and synthetic data can attenuate model collapse, though the proportion of real data must remain high to maintain model performance. Their findings support the idea that synthesized data alone cannot sustain model quality across iterations without a significant quantity of real data.

Gerstgrasser et al. (2024) argue that model collapse can be avoided entirely if data is accumulated rather than replaced across iterations. Their empirical studies on language models, diffusion models, and variational autoencoders indicate that accumulating both real and synthetic data helps maintain model performance over time, breaking the recursive degradation loop that leads to collapse.

The most relevant work to our study is Feng et al. (2024) where the authors examined the effects of synthesized data on model performance in a non-recursive setting, using the concept of reinforcement through feedback to select high quality synthetic data. Their theoretical results, based on Gaussian mixture models, showed that adding feedback significantly improves the robustness of models trained on synthesized data. However, their setup assumes that only labels, not features, are noisy. Additionally, their focus is primarily on scenarios where only the number of data points, n , grows to infinity. Other (practical) scenarios where for instance the feature dimension, p , grows at a fixed ratio with n are not covered.

Our work extends the Gaussian mixture model setup to include both noisy features and labels, which is a more realistic scenario when training on synthesized data. Additionally, we consider a high-dimensional regime where both p and n grow to infinity with a fixed ratio, a setup often used in Random Matrix Theory (RMT). This allows us to study the interaction between feature dimension, pruner error, and data size in a more comprehensive manner. Our approach also accounts for the presence of mixed data—original and synthetic—providing a more realistic framework for studying the effect of synthetic data in practical applications.

3 THEORETICAL SETUP

Real data. We suppose that real data consists of n p -dimensional i.i.d. vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ sampled from a Gaussian mixture of two distinct isotropic clusters $\mathcal{C}_1$ and $\mathcal{C}_2$ of means $\pm \boldsymbol{\mu}$ with $\boldsymbol{\mu} \in \mathbb{R}^p$. Essentially, for $a \in \{1, 2\}$, each data vector $\mathbf{x}_i \in \mathcal{C}_a$ has a corresponding label $y_i = (-1)^a$ and is sampled as

$$\mathbf{x}_i = y_i \boldsymbol{\mu} + \mathbf{z}_i, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p). \quad (1)$$

Generative model. To generate synthetic data, we consider the generative model corresponding to maximum likelihood which consists of estimating the underlying first and second-order statistics of the real data with their empirical estimates. In particular, we suppose that we are given a subset $\hat{n} \leq n$ of the real dataset $(\mathbf{x}_i, y_i)_{i=1}^{\hat{n}}$ on which we can estimate the statistics. This setup allows us to model a situation where new real data samples might be available to train next-generation models and the parameter $\hat{n}$ offers control over the generative model quality. The statistics for generating synthetic data are therefore computed using the following estimates

$$\hat{\boldsymbol{\mu}} = \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} y_i \mathbf{x}_i, \quad \hat{\mathbf{C}} = \frac{1}{\hat{n}} \sum_{i=1}^{\hat{n}} (y_i \mathbf{x}_i - \hat{\boldsymbol{\mu}})(y_i \mathbf{x}_i - \hat{\boldsymbol{\mu}})^\top. \quad (2)$$

Synthetic data. We consider that synthetic data is generated as m i.i.d. vectors $\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m \in \mathbb{R}^p$ with corresponding (noisy) labels $\tilde{y}_1, \dots, \tilde{y}_m = \pm 1$ such that $\tilde{\mathbf{x}}_i \in \tilde{\mathcal{C}}_a$ with true label $\tilde{y}_i = (-1)^a$ for $a \in \{1, 2\}$ is sampled as ($\tilde{\mathcal{C}}_1$ and $\tilde{\mathcal{C}}_2$ denote the synthetic clusters)

$$\tilde{\mathbf{x}}_i = \tilde{y}_i \hat{\boldsymbol{\mu}} + \hat{\mathbf{C}}^{\frac{1}{2}} \tilde{\mathbf{z}}_i, \quad \tilde{\mathbf{z}}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p), \quad (3)$$

and the labels $\tilde{y}_i$ are generated such that $\mathbb{P}\{\tilde{y}_i = \bar{y}_i\} = 1 - \varepsilon$ where $\varepsilon \geq 0$ controls label noise. Essentially, the **quality** of synthetic data depends on the **sample size** $\hat{n}$ and the **label noise rate** ε .

In the asymptotic regime where $\hat{n} \rightarrow \infty$ with $\frac{p}{\hat{n}} \rightarrow 0$, we can generate synthetic samples that follow asymptotically the exact same distribution as of the real ones, and therefore only label noise is relevant to the quality of the synthetic data. However, in the regime when both $\hat{n}, p \rightarrow \infty$ with $\frac{p}{\hat{n}} \rightarrow \hat{\eta} > 0$, while the estimation of $\boldsymbol{\mu}$ with $\hat{\boldsymbol{\mu}}$ remains consistent, the estimation of the covariance is not. In fact, in this regime $\|\hat{\mathbf{C}} - \mathbf{I}_p\| \not\rightarrow 0$ and the eigenvalues of $\hat{\mathbf{C}}$ spread in the vicinity of 1 which is described in the limit by the Marchenko-Pastur law (Marchenko & Pastur, 1967) as depicted in Fig. 1. Eventually, such inconsistency in estimating the second moment in high dimensions yields a distribution shift between synthetic and real data, which might cause a drop in performance when training a new model on synthetic data generated with $\hat{\boldsymbol{\mu}}$ and $\hat{\mathbf{C}}$. In the remainder, we describe precisely how the performance of a simple classifier is affected in this regime.

Objective. Our goal throughout the paper is to study the effect of synthetic data when training on a mixture of the n real and m synthetic data described above, i.e., with the following objective function

$$\mathcal{L}(\mathbf{w}) := \underbrace{\frac{1}{n+m} \sum_{i=1}^n \ell(\mathbf{x}_i, y_i; \mathbf{w})}_{\text{real data}} + \underbrace{\frac{1}{n+m} \sum_{i=1}^m q_i \ell(\tilde{\mathbf{x}}_i, \tilde{y}_i; \mathbf{w})}_{\text{synthetic data}}, \quad (4)$$

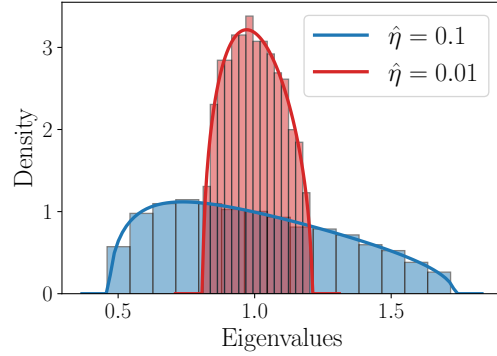


Figure 1: Illustration of the Marchenko-Pastur law: The histogram of eigenvalues of the empirical covariance matrix $\hat{\mathbf{C}}$ (as per equation (2)) using different values of $\hat{n}$. The histograms correspond to $p = 500$ and $\hat{n} = 500$ (in blue) and $\hat{n} = 5 \times 10^4$ (in red). The line plots depict the limiting Marchenko-Pastur law. As $\hat{n}$ grows, the distribution of eigenvalues shrinks towards 1.